



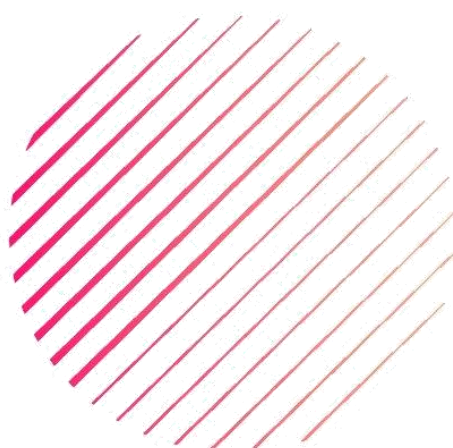
PROJECT REPORT

How Open-Source Digital Tools Are Opening Up the Frontier

Indigenous Young Women-Led Collective | Northern Kenya ASAL

Implementation Period: 2025

Locations: Turkana and Samburu Counties



1. Executive Summary

In 2024–2025, Beyond Borders piloted the use of open-source digital tools to address information and communication gaps in Northern Kenya’s pastoralist frontier. The project trained 20 indigenous young women to deploy, adapt, and locally govern open-source applications for mapping, storytelling, and secure communication — tools that are free to use, modifiable, and free from corporate data extraction.

The pilot demonstrated that open-source tools can bypass the proprietary lock-in and surveillance risks of commercial platforms, while remaining adaptable to low-connectivity, mobile pastoralist contexts. When combined with community governance and offline-first workflows, open-source becomes not just a technical choice but a political one: a refusal of digital colonialism and an assertion of local control over community data and knowledge.

20	4	35+	0
Young women trained	Open-source tools tested	Community records created	Corporate cloud uploads

2. Background and Rationale

Pastoralist communities in Northern Kenya have historically been “opened up” by external forces — colonial administrators, researchers, NGOs, and now technology corporations — each extracting data, labour, or knowledge while offering limited lasting benefit. Commercial digital platforms, even when nominally free, impose terms of service that few community members can read or negotiate, harvest behavioural data, and can terminate access without recourse.

Open-source software offers a structural alternative: code that is publicly inspectable, modifiable, and deployable without licensing fees or corporate control. However, open-source tools are rarely designed with ASAL realities in mind, and technical support is typically available only to those with internet access and English fluency. Beyond Borders designed this pilot to bridge that gap — training indigenous young women to become the local adapters and maintainers of open-source tools in frontier contexts.

3. Objectives

- Introduce and test open-source tools relevant to pastoralist community needs (mapping, archiving, communication).

- Train indigenous young women to install, configure, and troubleshoot open-source applications on local devices.
- Assess whether open-source tools can function reliably in low-connectivity, low-power ASAL conditions.
- Establish community governance protocols that keep open-source deployments under local control and free from external data extraction.
- Document the political and practical dimensions of choosing open-source in indigenous contexts.

4. Approach and Methods

4.1 Tool selection

Through community consultation, four categories of open-source tools were selected:

- **Mapeo** (offline collaborative mapping) for resource and territory documentation.
- **KoboToolbox** (offline data collection) for community surveys and oral history metadata.
- **Syncthing** (peer-to-peer file sync) for secure, device-to-device file transfer without cloud servers.
- **Jitsi Meet** (self-hosted video conferencing) for occasional inter-settlement meetings when connectivity permits.

4.2 Training

Twenty young women participated in hands-on workshops covering installation, offline configuration, basic troubleshooting, and the ethical rationale for open-source. Training emphasised that these tools belong to no corporation and can be modified to fit community needs.

4.3 Deployment and governance

Tools were deployed on local devices (tablets, phones, external drives) managed by the collective. Syncthing was configured for local mesh networks rather than internet-dependent sync. No data passed through commercial cloud services.

5. Key Results and Outputs

Tangible outputs

- 20 young women trained as open-source tool adapters and community support providers.
- 35+ community records (maps, oral history metadata, survey data) created using open-source platforms.
- Four fully offline-capable toolkits with ASAL-adapted user guides in local languages.
- A community policy committing Beyond Borders to open-source and local-data principles for all future digital work.

Relational and capacity outcomes

- Demonstration that indigenous women can manage complex software configuration without external engineers.
- Elimination of corporate data extraction from documented community knowledge workflows.
- Increased community understanding of software ownership and the politics of digital infrastructure.
- Local capacity to refuse proprietary platforms when they do not serve community interests.

6. Challenges and Lessons Learned

- **Technical documentation is exclusionary.** Most open-source guides assume English fluency and urban technical literacy. Local-language, visual, and oral training methods were essential adaptations.
- **Device limitations.** Some open-source tools require more processing power or memory than available on older shared phones. Lightweight alternatives must always be prioritised.
- **Occasional connectivity is still a challenge.** Even peer-to-peer sync requires occasional proximity or local network access. Purely manual transfer (SD cards, external drives) remained necessary for some settlements.
- **The “free” myth.** Open-source is free as in freedom, not always free as in cost. Local support, training, and maintenance require time and resources that must be funded.
- **Resistance from partners.** Some institutional partners preferred familiar proprietary platforms. Community open-source advocates needed negotiation skills to maintain collective standards.

Key lesson: *Open-source is not a magic solution, but it shifts power. When indigenous women control the tools, the frontier is opened on community terms — not by external extraction, but by local digital sovereignty.*

7. Contribution to Beyond Borders’ Work

This project established open-source and digital sovereignty as core pillars of Beyond Borders’ technology strategy. It proved that frontier communities need not accept the platforms offered by corporations or distant governments; they can build, adapt, and govern their own. The trained cohort now serves as internal technical support for all collective digital projects.